

The Notebooks • Era IV — Solve

Charcoal algebra folio

Testing Mode • UK • IB • US

Exercises

T.IV.5.001 • Surds GCSE Higher full toolkit pack • challenge • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{8}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.5.002 • Rationalise two-term denominators under time • core • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{20}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.5.003 • Surds in exact geometric answers • core • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{27}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.IV.5.004 • Fractional indices $\leftrightarrow$ radical form conversions • core • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{12}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.IV.5.005 • Solve equations with fractional indices • challenge • UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{72}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.5.006 • Nested radicals simplify (strange) • strange • UK_GCSE_H,IB_AA_HL

- (a) [never/neutral/1mk] Simplify $\sqrt{20}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{5}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{5}+2)(\sqrt{5}-2)$.

T.IV.5.007 • Simplify expressions with mixed surds and indices • challenge • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{50}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.IV.5.008 • Rationalise with conjugate under radicals • challenge • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{20}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.IV.5.009 • Surds non-calc paper segment 1 • challenge • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{20}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.IV.5.010 • Surds non-calc paper segment 2 • challenge • UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{32}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.IV.5.011 · Surds non-calc paper segment 3 · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{18}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.IV.7.001 · Completing the square exam method pack · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 8x + 15$.
- (b) [never/UK/2mk] Solve $x^2 - 8x + 15 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 10x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.002 · Turning points via completing the square · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 11x + 28$.
- (b) [never/UK/2mk] Solve $x^2 - 11x + 28 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 8x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.003 · Sketch quadratics from completed square · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 10x + 24$.
- (b) [never/UK/2mk] Solve $x^2 - 10x + 24 = 0$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Sketch quadratics from completed square»: vary one parameter, capture before/after, write one observation.

T.IV.7.004 · Quadratic formula exact surd answers · core · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{32}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{5}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{5}+2)(\sqrt{5}-2)$.

T.IV.7.005 · Discriminant: number of roots exam stems · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 8x + 15$.
- (b) [never/UK/2mk] Solve $x^2 - 8x + 15 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 6x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.006 · Form quadratic equations from roots/context · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 6x + 8$.
- (b) [never/UK/2mk] Solve $x^2 - 6x + 8 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 4x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.007 · Quadratic simultaneous with linear · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 12x + 35$.
- (b) [never/UK/2mk] Solve $x^2 - 12x + 35 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 10x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.008 · Iteration: rearrange to $x=g(x)$ form · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 7x + 6$.
- (b) [never/UK/2mk] Solve $x^2 - 7x + 6 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.009 · Iteration: show a root lies in an interval · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 5x + 4$.
- (b) [never/UK/2mk] Solve $x^2 - 5x + 4 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.010 · Iteration: perform and interpret convergence · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 3x + 2$.
- (b) [never/UK/2mk] Solve $x^2 - 3x + 2 = 0$.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Iteration: perform and interpret convergence»: record menu path and numerical output.

T.IV.7.011 · Quadratic inequalities critical values method · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 10x + 24$.
- (b) [never/UK/2mk] Solve $x^2 - 10x + 24 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 8x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.012 · Projectile worded GCSE/IB pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 5x + 6$.
- (b) [never/UK/2mk] Solve $x^2 - 5x + 6 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 4x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.013 · Hidden quadratics (exp/trig substitutions) · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 3x + 2$.
- (b) [never/UK/2mk] Solve $x^2 - 3x + 2 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.014 · Quadratic model fitting with GDC · core · IB_AI_SL,IB_AI_HL

- (a) [never/neutral/2mk] Factorise $x^2 - 8x + 7$.
- (b) [never/IB/2mk] Solve $x^2 - 8x + 7 = 0$.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Quadratic model fitting with GDC»: record menu path and numerical output.

T.IV.7.015 · Sign-change decimal search before iteration · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 9x + 18$.
- (b) [never/UK/2mk] Solve $x^2 - 9x + 18 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 6x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.016 · Intersecting chord/quadratic geometry link · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 10x + 24$.
- (b) [never/UK/2mk] Solve $x^2 - 10x + 24 = 0$.
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Intersecting chord/quadratic geometry link»: build the figure, measure, verify the claimed relation.

T.IV.7.017 · Completing square exam battery 1 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 4x + 3$.
- (b) [never/UK/2mk] Solve $x^2 - 4x + 3 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.018 · Completing square exam battery 2 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 6x + 5$.
- (b) [never/UK/2mk] Solve $x^2 - 6x + 5 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.019 · Completing square exam battery 3 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 9x + 14$.
- (b) [never/UK/2mk] Solve $x^2 - 9x + 14 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 4x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.020 · Completing square exam battery 4 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 7x + 10$.
- (b) [never/UK/2mk] Solve $x^2 - 7x + 10 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 10x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.021 · Iteration numerical table 1 · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 9x + 20$.
- (b) [never/UK/2mk] Solve $x^2 - 9x + 20 = 0$.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Iteration numerical table 1»: record menu path and numerical output.

T.IV.7.022 · Iteration numerical table 2 · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 11x + 28$.
- (b) [never/UK/2mk] Solve $x^2 - 11x + 28 = 0$.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Iteration numerical table 2»: record menu path and numerical output.

T.IV.7.023 · Iteration numerical table 3 · core · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 7x + 10$.
- (b) [never/UK/2mk] Solve $x^2 - 7x + 10 = 0$.
- (c) [gdc/UK/3mk ■ LAB] GDC lab for «Iteration numerical table 3»: record menu path and numerical output.

T.IV.1.001 · Inequalities regions on graphs (linear) · core · UK_GCSE_H

(a) [never/neutral/1mk] Solve $7x + 7 = 56$.

(b) [never/UK/2mk] Solve $7(x - 7) = 7$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Inequalities regions on graphs (linear)»: vary one parameter, capture before/after, write one observation.

T.IV.1.002 · Set notation for solution sets (UK) · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Solve $7x + 7 = 28$.

(b) [never/UK/2mk] Solve $7(x - 7) = -21$.

(c) [never/UK/2mk] List integers n satisfying $3 < n \leq 7$.

T.IV.1.003 · Exact vs approximate solutions language · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Simplify $\sqrt{12}$.

(b) [never/UK/1mk] Rationalise $1/\sqrt{3}$.

(c) [never/UK/2mk] Expand and simplify $(\sqrt{3}+2)(\sqrt{3}-2)$.

T.IV.2.001 · Function notation $f(x)$ exam fluency · core · UK_GCSE_F,UK_KS3,IB_MYP

(a) [never/neutral/1mk] Expand $6(x+6)$.

(b) [never/UK/2mk] Expand $(x+1)(x+6)$.

(c) [never/UK/2mk] Factorise $x^2 + 7x + 6$.

T.IV.2.002 · Domain and range from graphs and equations · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $7(x+3)$.

(b) [never/UK/2mk] Expand $(x+3)(x+6)$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Domain and range from graphs and equations»: vary one parameter, capture before/after, write one observation.

T.IV.2.003 · Composite functions fog and gof · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $7(x+4)$.

(b) [never/UK/2mk] Expand $(x+5)(x+1)$.

(c) [never/UK/2mk] Factorise $x^2 + 6x + 5$.

T.IV.2.004 · Inverse functions algebraically · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $2(x+5)$.

(b) [never/UK/2mk] Expand $(x+1)(x+1)$.

(c) [never/UK/2mk] Factorise $x^2 + 2x + 1$.

T.IV.2.005 · Piecewise functions interpret and graph · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

(a) [never/neutral/1mk] Expand $4(x+2)$.

(b) [never/US/2mk] Expand $(x+2)(x+4)$.

(c) [allowed/US/3mk ■ LAB] Desmos lab for «Piecewise functions interpret and graph»: vary one parameter, capture before/after, write one observation.

T.IV.2.006 · Transformations of graphs $y=f(x)$ exam pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $6(x+1)$.

(b) [never/UK/2mk] Expand $(x+5)(x+6)$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Transformations of graphs $y=f(x)$ exam pack»: vary one parameter, capture before/after, write one observation.

T.IV.2.007 · Modulus function graphs and equations · challenge · UK_GCSE_H,IB_AA_SL

(a) [never/neutral/1mk] Expand $4(x+5)$.

(b) [never/UK/2mk] Expand $(x+1)(x+3)$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Modulus function graphs and equations»: vary one parameter, capture before/after, write one observation.

T.IV.2.008 · Vertical line test and function definition exam · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

(a) [never/neutral/1mk] Expand $5(x+5)$.

(b) [never/US/2mk] Expand $(x+2)(x+4)$.

(c) [allowed/US/3mk ■ LAB] Desmos lab for «Vertical line test and function definition exam»: vary one parameter, capture before/after, write one observation.

T.IV.2.009 · 1–1 and onto language for inverses · challenge · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Expand $6(x+1)$.
- (b) [never/IB/2mk] Expand $(x+4)(x+3)$.
- (c) [never/IB/2mk] Factorise $x^2 + 7x + 12$.

T.IV.2.010 · Parametric definition of functions intro · core · IB_AA_SL,IB_AA_HL,AP_CALC_BC

- (a) [never/neutral/1mk] Expand $5(x+1)$.
- (b) [never/IB/2mk] Expand $(x+5)(x+2)$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Parametric definition of functions intro»: vary one parameter, capture before/after, write one observation.

T.IV.2.011 · Self-inverse functions recognition · challenge · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/1mk] Expand $4(x+1)$.
- (b) [never/IB/2mk] Expand $(x+3)(x+3)$.
- (c) [never/IB/2mk] Factorise $x^2 + 6x + 9$.

T.IV.3.001 · Perpendicular bisector equation · core · UK_GCSE_H

- (a) [never/neutral/1mk] [Perpendicular bisector equation] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 3 \times 2$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.IV.3.002 · Distance from point to line formula use · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Distance from point to line formula use] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $9 \times 3 \times 5$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 9) standing on its base.

T.IV.3.003 · Equation of circle centre origin + tangent (GCSE H) · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] [Equation of circle centre origin + tangent (GCSE H)] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $4 \times 6 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/3mk ■ LAB] Geometry construction lab for «Equation of circle centre origin + tangent (GCSE H)»: build the figure, measure, verify the claimed relation.

T.IV.3.004 · Equation of circle general; find centre/radius · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] [Equation of circle general; find centre/radius] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $3 \times 3 \times 3$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 3) standing on its base.

T.IV.4.001 · Simultaneous three equations intro · challenge · IB_AA_HL,IB_AI_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/IB/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$?

T.IV.4.002 · Non-linear simultaneous exam pack · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}\right)$?
- (b) [never/UK/1mk] $\begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/UK/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 1 & 1 \\ 4 & 3 \end{bmatrix}$?

T.IV.4.003 · Matrices 2×2 multiply and inverse (GCSE+) · challenge · UK_GCSE_H,IB_AA_HL,IB_AI_HL

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix}\right)$?
- (b) [never/UK/1mk] $\begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/UK/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 3 & 1 \\ 5 & 3 \end{bmatrix}$?

T.IV.4.004 · Matrix transformations of the plane intro · challenge · IB_AI_HL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}\right)$?
- (b) [never/IB/1mk] $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \times \mathbf{I}$?
- (c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Matrix transformations of the plane intro»: build the figure, measure, verify the claimed relation.

T.IV.4.005 · Graphical linear programming light (regions) · strange · IB_AI_HL,UK_GCSE_H

(a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix}\right)$?

(b) [never/IB/1mk] $\begin{bmatrix} 1 & 4 \\ 1 & 3 \end{bmatrix} \times \mathbf{I}$?

(c) [allowed/IB/3mk ■ LAB] Desmos lab for «Graphical linear programming light (regions)»: vary one parameter, capture before/after, write one observation.

T.IV.6.001 · Factor theorem exam use · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $7(x+3)$.

(b) [never/UK/2mk] Expand $(x+1)(x+1)$.

(c) [never/UK/2mk] Factorise $x^2 + 2x + 1$.

T.IV.6.002 · Remainder theorem exam use · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $2(x+4)$.

(b) [never/UK/2mk] Expand $(x+5)(x+1)$.

(c) [never/UK/2mk] Factorise $x^2 + 6x + 5$.

T.IV.6.003 · Algebraic division of cubics · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $7(x+2)$.

(b) [never/UK/2mk] Expand $(x+1)(x+4)$.

(c) [never/UK/2mk] Factorise $x^2 + 5x + 4$.

T.IV.6.004 · Sketch cubic and quartic from factorised form · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $3(x+3)$.

(b) [never/UK/2mk] Expand $(x+5)(x+4)$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Sketch cubic and quartic from factorised form»: vary one parameter, capture before/after, write one observation.

T.IV.6.005 · Expand and factorise mixed higher exam set · core · UK_GCSE_H

(a) [never/neutral/1mk] Expand $7(x+1)$.

(b) [never/UK/2mk] Expand $(x+5)(x+4)$.

(c) [never/UK/2mk] Factorise $x^2 + 9x + 20$.

T.IV.6.006 · Binomial expansion positive integer n · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $3(x+5)$.

(b) [never/UK/2mk] Expand $(x+4)(x+1)$.

(c) [never/UK/2mk] Factorise $x^2 + 5x + 4$.

T.IV.6.007 · Binomial approximation $(1+x)^n$ for small x · challenge · IB_AA_HL,UK_GCSE_H

(a) [never/neutral/1mk] Expand $4(x+4)$.

(b) [never/IB/2mk] Expand $(x+4)(x+4)$.

(c) [never/IB/2mk] Factorise $x^2 + 8x + 16$.

T.IV.6.008 · Sketch from leading term and roots only · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

(a) [never/neutral/1mk] Expand $5(x+4)$.

(b) [never/UK/2mk] Expand $(x+3)(x+4)$.

(c) [allowed/UK/3mk ■ LAB] Desmos lab for «Sketch from leading term and roots only»: vary one parameter, capture before/after, write one observation.

T.IV.8.001 · Argand diagram exam plotting · core · IB_AA_HL,IB_AA_SL

(a) [never/neutral/1mk] Expand $3(x+6)$.

(b) [never/IB/2mk] Expand $(x+1)(x+4)$.

(c) [allowed/IB/3mk ■ LAB] Geometry construction lab for «Argand diagram exam plotting»: build the figure, measure, verify the claimed relation.

T.IV.8.002 · Modulus-argument form conversions · core · IB_AA_HL

(a) [never/neutral/1mk] Expand $4(x+6)$.

(b) [never/IB/2mk] Expand $(x+1)(x+5)$.

(c) [never/IB/2mk] Factorise $x^2 + 6x + 5$.

T.IV.8.003 · De Moivre exam applications · challenge · IB_AA_HL

- (a) [never/neutral/1mk] Expand $5(x+4)$.
- (b) [never/IB/2mk] Expand $(x+5)(x+4)$.
- (c) [never/IB/2mk] Factorise $x^2 + 9x + 20$.

T.IV.8.004 · Complex loci $|z-a|=r$ and $\arg(z-a)=\theta$ · challenge · IB_AA_HL

- (a) [never/neutral/1mk] Expand $7(x+4)$.
- (b) [never/IB/2mk] Expand $(x+3)(x+1)$.
- (c) [never/IB/2mk] Factorise $x^2 + 4x + 3$.

T.IV.8.005 · Solve equations in complex numbers · challenge · IB_AA_HL

- (a) [never/neutral/1mk] Expand $6(x+6)$.
- (b) [never/IB/2mk] Expand $(x+2)(x+3)$.
- (c) [never/IB/2mk] Factorise $x^2 + 5x + 6$.

T.IV.9.001 · Even/odd functions recognition · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Expand $2(x+4)$.
- (b) [never/US/2mk] Expand $(x+4)(x+6)$.
- (c) [never/US/2mk] Factorise $x^2 + 10x + 24$.

T.IV.9.002 · Absolute value inequalities · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Expand $4(x+4)$.
- (b) [never/US/2mk] Expand $(x+5)(x+2)$.
- (c) [never/US/2mk] Factorise $x^2 + 7x + 10$.

T.IV.9.003 · Asymptotes of rational functions exam · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $5(x+2)$.
- (b) [never/UK/2mk] Expand $(x+4)(x+2)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Asymptotes of rational functions exam»: vary one parameter, capture before/after, write one observation.

T.IV.9.004 · Combined transformations order matters · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $2(x+5)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+4)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Combined transformations order matters»: vary one parameter, capture before/after, write one observation.

T.IV.9.005 · Step functions and floor in modelling (light) · strange · IB_AI_SL,IB_AI_HL

- (a) [never/neutral/1mk] Expand $3(x+3)$.
- (b) [never/IB/2mk] Expand $(x+2)(x+5)$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Step functions and floor in modelling (light)»: vary one parameter, capture before/after, write one observation.

T.IV.9.006 · Oblique asymptotes by division · challenge · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Expand $6(x+2)$.
- (b) [never/US/2mk] Expand $(x+3)(x+4)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Oblique asymptotes by division»: vary one parameter, capture before/after, write one observation.

T.IV.10.001 · Polynomial inequalities on number line · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Solve $4x + 7 = 43$.
- (b) [never/US/2mk] Solve $4(x - 7) = 12$.
- (c) [never/US/2mk] List integers n satisfying $9 < n \leq 13$.

T.IV.11.001 · Solve rational equations check extraneous · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Solve $8x + 7 = 31$.
- (b) [never/UK/2mk] Solve $8(x - 7) = -24$.
- (c) [never/UK/2mk] List integers n satisfying $3 < n \leq 7$.

T.IV.11.002 · Radical equations isolate and check · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{50}$.
- (b) [never/US/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/US/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.11.003 · Graphical solution of radical/rational eqs · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{75}$.
- (b) [never/US/1mk] Rationalise $1/\sqrt{3}$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Graphical solution of radical/rational eqs»: vary one parameter, capture before/after, write one observation.

T.IV.12.001 · Compound interest continuous ($A=Pe^{rt}$) · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [allowed/neutral/1mk] Increase £120 by 5%.
- (b) [allowed/US/2mk] After a 5% increase a price is £126. Find the original.
- (c) [allowed/US/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.IV.12.002 · GCSE growth and decay worded pack · core · UK_GCSE_H

- (a) [allowed/neutral/1mk] Increase £120 by 10%.
- (b) [allowed/UK/2mk] After a 10% increase a price is £132. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 10% then by another 10%. Overall % increase?

T.IV.12.003 · IB GDC exponential regression workflow · core · IB_AI_SL,IB_AI_HL,IB_AA_SL

- (a) [gdc/neutral/1mk] Increase £120 by 12%.
- (b) [gdc/IB/2mk] After a 12% increase a price is £134. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «IB GDC exponential regression workflow»: record menu path and numerical output.

T.IV.12.004 · Log scales interpretation · core · IB_AI_SL,IB_AI_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Increase £120 by 10%.
- (b) [allowed/IB/2mk] After a 10% increase a price is £132. Find the original.
- (c) [allowed/IB/3mk ■ LAB] Graph lab for «Log scales interpretation»: plot, mark key points, state the trend.

T.IV.12.005 · Solve exp/log equations exam mixed pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £250 by 20%.
- (b) [allowed/UK/2mk] After a 20% increase a price is £300. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 20% then by another 20%. Overall % increase?

T.IV.12.006 · Change of base in exam calculations · core · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [gdc/neutral/1mk] Increase £250 by 15%.
- (b) [gdc/IB/2mk] After a 15% increase a price is £287. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Change of base in exam calculations»: record menu path and numerical output.

T.IV.12.007 · Half-life multi-step worded · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £200 by 25%.
- (b) [allowed/UK/2mk] After a 25% increase a price is £250. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 25% then by another 25%. Overall % increase?

T.IV.12.008 · Modelling cycle: formulate–solve–interpret–evaluate · challenge · IB_AI_SL,IB_AI_HL,IB_AI_HL

- (a) [allowed/neutral/1mk] Increase £80 by 12%.
- (b) [allowed/IB/2mk] After a 12% increase a price is £89. Find the original.
- (c) [allowed/IB/3mk] A quantity rises by 12% then by another 12%. Overall % increase?

T.IV.12.009 · Compare linear vs exponential models · challenge · IB_AI_SL,IB_AI_HL,US_CCSS_HS

- (a) [gdc/neutral/1mk] Increase £250 by 20%.
- (b) [gdc/IB/2mk] After a 20% increase a price is £300. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Compare linear vs exponential models»: record menu path and numerical output.

T.IV.12.010 · Logarithmic linearisation of data · challenge · IB_AI_SL,IB_AI_HL,IB_AI_HL

- (a) [gdc/neutral/1mk] Increase £120 by 15%.
- (b) [gdc/IB/2mk] After a 15% increase a price is £138. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Logarithmic linearisation of data»: record menu path and numerical output.

T.IV.12.011 · Newton's law of cooling discrete steps · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £400 by 5%.
- (b) [allowed/UK/2mk] After a 5% increase a price is £420. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.IV.12.012 · Effective annual rate vs nominal · challenge · IB_AI_SL,IB_AI_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Increase £250 by 20%.
- (b) [allowed/IB/2mk] After a 20% increase a price is £300. Find the original.
- (c) [allowed/IB/3mk] A quantity rises by 20% then by another 20%. Overall % increase?

T.IV.12.013 · Exp/log modelling context 1 · challenge · IB_AI_SL,IB_AI_HL,UK_GCSE_H

- (a) [gdc/neutral/1mk] Increase £400 by 12%.
- (b) [gdc/IB/2mk] After a 12% increase a price is £448. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Exp/log modelling context 1»: record menu path and numerical output.

T.IV.12.014 · Exp/log modelling context 2 · challenge · IB_AI_SL,IB_AI_HL,UK_GCSE_H

- (a) [gdc/neutral/1mk] Increase £250 by 5%.
- (b) [gdc/IB/2mk] After a 5% increase a price is £262. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Exp/log modelling context 2»: record menu path and numerical output.

T.IV.12.015 · Exp/log modelling context 3 · challenge · IB_AI_SL,IB_AI_HL,UK_GCSE_H

- (a) [gdc/neutral/1mk] Increase £250 by 25%.
- (b) [gdc/IB/2mk] After a 25% increase a price is £312. Find the original.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Exp/log modelling context 3»: record menu path and numerical output.

T.IV.13.001 · nth term of linear sequences (GCSE) · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/2mk] nth term of 7, 13, 19, 25, ...
- (b) [never/UK/1mk] 20th term: $a=7, d=6$.
- (c) [never/UK/2mk] S_n for $a=7, d=6$.

T.IV.13.002 · Quadratic sequences nth term · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 2, 4, 6, 8, ...
- (b) [never/UK/1mk] 20th term: $a=2, d=2$.
- (c) [never/UK/2mk] S_n for $a=2, d=2$.

T.IV.13.003 · Geometric sequences GCSE/IB worded · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 5, 11, 17, 23, ...
- (b) [never/UK/1mk] 20th term: $a=5, d=6$.
- (c) [never/UK/2mk] S_n for $a=5, d=6$.

T.IV.13.004 · Arithmetic series exam applications · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 9, 15, 21, 27, ...
- (b) [never/UK/1mk] 20th term: $a=9, d=6$.
- (c) [never/UK/2mk] S_n for $a=9, d=6$.

T.IV.13.005 · Infinite geometric series $|r|<1$ condition · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 4, 10, 16, 22, ...
- (b) [never/UK/1mk] 20th term: $a=4, d=6$.
- (c) [never/UK/2mk] S_n for $a=4, d=6$.

T.IV.13.006 · Sigma notation exam fluency · core · IB_AA_SL,IB_AA_HL,UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 9, 14, 19, 24, ...
- (b) [never/IB/1mk] 20th term: $a=9, d=5$.
- (c) [never/IB/2mk] S_n for $a=9, d=5$.

T.IV.13.007 · Recurrence relations $x_{n+1}=ax_n+b$ · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 7, 12, 17, 22, ...
- (b) [never/UK/1mk] 20th term: $a=7, d=5$.
- (c) [never/UK/2mk] S_n for $a=7, d=5$.

T.IV.13.008 · Fibonacci-type sequences properties · strange · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 6, 12, 18, 24, ...
- (b) [never/UK/1mk] 20th term: $a=6$, $d=6$.
- (c) [never/UK/2mk] S_n for $a=6$, $d=6$.

T.IV.13.009 · Sequences modelling IB AI contexts · core · IB_AI_SL,IB_AI_HL

- (a) [never/neutral/2mk] nth term of 7, 11, 15, 19, ...
- (b) [never/IB/1mk] 20th term: $a=7$, $d=4$.
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Sequences modelling IB AI contexts»: record menu path and numerical output.

T.IV.13.010 · Prove a sequence is arithmetic/geometric · challenge · IB_AA_SL,IB_AA_HL

- (a) [never/neutral/2mk] nth term of 4, 11, 18, 25, ...
- (b) [never/IB/1mk] 20th term: $a=4$, $d=7$.
- (c) [never/IB/2mk] S_n for $a=4$, $d=7$.

T.IV.13.011 · Financial series: annuities simple · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 5, 7, 9, 11, ...
- (b) [never/UK/1mk] 20th term: $a=5$, $d=2$.
- (c) [never/UK/2mk] S_n for $a=5$, $d=2$.

T.IV.13.012 · IB AA HL induction on sequences preview · challenge · IB_AA_HL

- (a) [never/neutral/2mk] List the three structural steps of mathematical induction.
- (b) [never/IB/3mk] Inductive step only: assume $1+\dots+k=k(k+1)/2$; prove for $k+1$.
- (c) [never/IB/2mk] Name one write-up error that loses IB AA HL induction marks.

T.IV.13.013 · Compare arithmetic vs geometric growth stories · core · IB_AI_SL,IB_AI_HL,UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 6, 13, 20, 27, ...
- (b) [never/IB/1mk] 20th term: $a=6$, $d=7$.
- (c) [never/IB/2mk] S_n for $a=6$, $d=7$.

T.IV.13.014 · Sequences GCSE mixed paper 1 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 2, 4, 6, 8, ...
- (b) [never/UK/1mk] 20th term: $a=2$, $d=2$.
- (c) [never/UK/2mk] S_n for $a=2$, $d=2$.

T.IV.13.015 · Sequences GCSE mixed paper 2 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 3, 5, 7, 9, ...
- (b) [never/UK/1mk] 20th term: $a=3$, $d=2$.
- (c) [never/UK/2mk] S_n for $a=3$, $d=2$.

T.IV.13.016 · Sequences GCSE mixed paper 3 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 2, 6, 10, 14, ...
- (b) [never/UK/1mk] 20th term: $a=2$, $d=4$.
- (c) [never/UK/2mk] S_n for $a=2$, $d=4$.

T.IV.13.017 · Sequences GCSE mixed paper 4 · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] nth term of 3, 10, 17, 24, ...
- (b) [never/UK/1mk] 20th term: $a=3$, $d=7$.
- (c) [never/UK/2mk] S_n for $a=3$, $d=7$.

T.IV.14.001 · Product rule and permutations exam pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 5 red, 2 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.002 · Combinations in probability contexts · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 5 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.003 · Conditional probability tree diagrams (GCSE H) · core · UK_GCSE_H

- (a) [never/neutral/1mk] Bag: 4 red, 5 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.004 · Independent vs mutually exclusive exam traps · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 7 red, 6 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.005 · Venn $\leftrightarrow$ tree $\leftrightarrow$ two-way translation · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 6 red, 2 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.006 · Expected value decision problems (S-MD) · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Bag: 7 red, 4 blue. $P(\text{red})$?
- (b) [never/US/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/US/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.007 · Bayes with medical/test contexts · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 8 red, 4 blue. $P(\text{red})$?
- (b) [never/US/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/US/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.008 · Inclusive exclusive counting with Venn · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 4 red, 3 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.009 · Conditional probability given complementary info · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 7 red, 6 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.010 · At least one complement strategy pack · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 4 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.011 · Law of large numbers simulation demo · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 8 red, 6 blue. $P(\text{red})$?
- (b) [never/US/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/US/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.012 · Rare event rule intuition · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 8 red, 2 blue. $P(\text{red})$?
- (b) [never/US/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/US/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.013 · Expected value of games fair/unfair · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 7 red, 7 blue. $P(\text{red})$?
- (b) [never/US/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/US/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.014 · Conditional probability medical screening FRQ · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Conditional probability medical screening FRQ»?
- (b) [allowed/US/2mk] Write a justification sentence frame for «Conditional probability medical screening FRQ»: Because ____, by ____, therefore ____.
- (c) [allowed/US/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability medical screening FRQ», then interpret.

T.IV.14.015 · mutually exclusive addition rule drills · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/1mk] Bag: 7 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.016 · Tree diagrams with three stages · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 8 red, 5 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.017 · Conditional probability FRQ-style 1 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 1»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Conditional probability FRQ-style 1»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 1», then interpret.

T.IV.14.018 · Conditional probability FRQ-style 2 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 2»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Conditional probability FRQ-style 2»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 2», then interpret.

T.IV.14.019 · Conditional probability FRQ-style 3 · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] AP/IB style: Which theorem or definition is central to «Conditional probability FRQ-style 3»?
- (b) [allowed/UK/2mk] Write a justification sentence frame for «Conditional probability FRQ-style 3»: Because ____, by ____, therefore ____.
- (c) [allowed/UK/3mk] Mini-FRQ: Give a 2-mark numeric or symbolic check that demonstrates «Conditional probability FRQ-style 3», then interpret.

T.IV.15.001 · Normal distribution empirical rule exam · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.15.002 · z-score calculations and interpretation · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.003 · Binomial probability GDC workflow · core · IB_AI_SL,IB_AI_HL,AP_STATS

- (a) [gdc/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [gdc/IB/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/IB/3mk ■ LAB] GDC lab for «Binomial probability GDC workflow»: record menu path and numerical output.

T.IV.15.004 · Scatter plots: describe strength/direction/form · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.15.005 · Correlation vs causation exam stems · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.15.006 · Least-squares residual interpretation · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Least-squares residual interpretation»: record menu path and numerical output.

T.IV.15.007 · Outliers effect on mean/median/regression · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.008 · Comparing distributions with multiple measures · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.IV.15.009 · Percentiles and cumulative relative frequency · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.010 · Skew interpretation from histograms/boxplots · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.IV.15.011 · Binomial vs geometric settings recognition · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.15.012 · Geometric distribution probability · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.013 · Normal probability GDC invNorm/normalcdf · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [gdc/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Normal probability GDC invNorm/normalcdf»: record menu path and numerical output.

T.IV.15.014 · Assessing normality: plots and rules · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Assessing normality: plots and rules»: record menu path and numerical output.

T.IV.15.015 · Transforming data: effect on mean/SD · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.15.016 · Residual plots: pattern means non-linear · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Residual plots: pattern means non-linear»: record menu path and numerical output.

T.IV.15.017 · Combining random variables mean/variance · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.018 · Linear regression prediction vs extrapolation · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.IV.15.019 · Discrete RV probability distribution tables · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.15.020 · Inverse normal cut-points for percentiles · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [gdc/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [gdc/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/3mk ■ LAB] GDC lab for «Inverse normal cut-points for percentiles»: record menu path and numerical output.

T.IV.15.021 · Effect of coding data (subtract/divide) · challenge · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(100, 15^2)$. Approx $P(85 < X < 115)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=130$ ($\mu=100$, $\sigma=15$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 115)$ for $X \sim N(100, 15^2)$.

T.IV.5.012 · Surds in exact trig non-calc hybrids · challenge · UK_GCSE_H

- (a) [never/neutral/1mk] Simplify $\sqrt{8}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.5.013 · Index equations leading to quadratics · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{18}$.
- (b) [never/UK/1mk] Rationalise $1/\sqrt{2}$.
- (c) [never/UK/2mk] Expand and simplify $(\sqrt{2}+2)(\sqrt{2}-2)$.

T.IV.7.024 · Completing square with $a \neq 1$ fluency · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 8x + 15$.
- (b) [never/UK/2mk] Solve $x^2 - 8x + 15 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 6x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.7.025 · Iteration: cobweb diagram interpretation · strange · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 9x + 20$.
- (b) [never/UK/2mk] Solve $x^2 - 9x + 20 = 0$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Iteration: cobweb diagram interpretation»: vary one parameter, capture before/after, write one observation.

T.IV.7.026 · Quadratic inequalities on number line shading · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] Factorise $x^2 - 8x + 15$.
- (b) [never/UK/2mk] Solve $x^2 - 8x + 15 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 10x + 7 = (x-p)^2 + q$. Find p,q.

T.IV.2.012 · Restrict domain to make inverse · challenge · IB_AA_SL,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] Expand $3(x+2)$.
- (b) [never/IB/2mk] Expand $(x+4)(x+5)$.
- (c) [allowed/IB/3mk ■ LAB] Desmos lab for «Restrict domain to make inverse»: vary one parameter, capture before/after, write one observation.

T.IV.2.013 · Function arithmetic (sum product quotient) · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Expand $4(x+4)$.
- (b) [never/US/2mk] Expand $(x+1)(x+5)$.
- (c) [allowed/US/3mk ■ LAB] Desmos lab for «Function arithmetic (sum product quotient)»: vary one parameter, capture before/after, write one observation.

T.IV.6.009 · Factorise cubics fully after one root · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $5(x+5)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+1)$.
- (c) [never/UK/2mk] Factorise $x^2 + 2x + 1$.

T.IV.6.010 · Polynomial division with remainders interpreted · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Expand $7(x+2)$.
- (b) [never/UK/2mk] Expand $(x+2)(x+6)$.
- (c) [never/UK/2mk] Factorise $x^2 + 8x + 12$.

T.IV.12.016 · Compound interest vs simple interest compare · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £250 by 15%.
- (b) [allowed/UK/2mk] After a 15% increase a price is £287. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 15% then by another 15%. Overall % increase?

T.IV.12.017 · Solve for time in exponential models · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [allowed/neutral/1mk] Increase £400 by 20%.
- (b) [allowed/UK/2mk] After a 20% increase a price is £480. Find the original.
- (c) [allowed/UK/3mk] A quantity rises by 20% then by another 20%. Overall % increase?

T.IV.13.018 · nth term from practical sequences contexts · core · UK_GCSE_F,UK_KS3,IB_MYP

- (a) [never/neutral/2mk] nth term of 5, 11, 17, 23, ...
- (b) [never/UK/1mk] 20th term: $a=5$, $d=6$.
- (c) [never/UK/2mk] S_n for $a=5$, $d=6$.

T.IV.13.019 · Sum of first n naturals / odds identities · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/2mk] nth term of 6, 12, 18, 24, ...
- (b) [never/UK/1mk] 20th term: $a=6$, $d=6$.
- (c) [never/UK/2mk] S_n for $a=6$, $d=6$.

T.IV.14.020 · Conditional probability given restricted sample space · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 6 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.14.021 · Counting with restrictions (no two adjacent) · challenge · UK_GCSE_H,IB_AA_SL,IB_AA_HL

- (a) [never/neutral/1mk] Bag: 4 red, 4 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.15.022 · Empirical rule vs Chebyshev light contrast · strange · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50$, $\sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.IV.15.023 · Binomial mean np and SD $\sqrt{(npq)}$ fluency · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(70, 10^2)$. Approx $P(60 < X < 80)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=90$ ($\mu=70$, $\sigma=10$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 80)$ for $X \sim N(70, 10^2)$.

T.IV.4.006 · Determinant condition for unique solution 2×2 · core · UK_GCSE_H,IB_AA_HL,US_CCSS_HS

- (a) [never/neutral/1mk] $\det\left(\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix}\right)$?
- (b) [never/UK/1mk] $\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix} \times \mathbf{I}$?
- (c) [never/UK/3mk] If $\det \neq 0$, inverse of $\begin{bmatrix} 4 & 5 \\ 4 & 5 \end{bmatrix}$?

T.IV.8.006 · Powers of i cycle fluency · core · IB_AA_HL,IB_AA_SL

- (a) [never/neutral/1mk] Expand $6(x+3)$.
- (b) [never/IB/2mk] Expand $(x+5)(x+6)$.
- (c) [never/IB/2mk] Factorise $x^2 + 11x + 30$.

T.IV.9.007 · Graph $y=1/f(x)$ from $y=f(x)$ features · challenge · UK_GCSE_H,IB_AA_SL,IB_AA_HL

- (a) [never/neutral/1mk] Expand $2(x+4)$.
- (b) [never/UK/2mk] Expand $(x+1)(x+1)$.
- (c) [allowed/UK/3mk ■ LAB] Desmos lab for «Graph $y=1/f(x)$ from $y=f(x)$ features»: vary one parameter, capture before/after, write one observation.

T.IV.3.005 · Find intersection of two circles algebraically · challenge · UK_GCSE_H,US_CCSS_HS,UK_GCSE_H

- (a) [never/neutral/1mk] [Find intersection of two circles algebraically] Which view is the plan — front, side, or top?
- (b) [never/UK/1mk] Cuboid $5 \times 3 \times 4$: how many rectangles in a correct net?
- (c) [allowed/UK/2mk] Describe the front elevation of a square-based pyramid (base 5) standing on its base.

T.IV.1.004 · Inequalities with fractions clear carefully · challenge · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Solve $2x + 6 = 8$.
- (b) [never/UK/2mk] Solve $2(x - 6) = -8$.
- (c) [never/UK/2mk] List integers n satisfying $1 < n \leq 5$.

T.IV.11.004 · Domain of radical and rational combined · core · US_CCSS_HS,UK_GCSE_H,IB_AA_SL

- (a) [never/neutral/1mk] Simplify $\sqrt{27}$.
- (b) [never/US/1mk] Rationalise $1/\sqrt{7}$.
- (c) [never/US/2mk] Expand and simplify $(\sqrt{7}+2)(\sqrt{7}-2)$.

T.IV.12.018 · Doubling time from continuous and discrete models · challenge · IB_AI_SL,IB_AI_HL,US_CCSS_HS

- (a) [allowed/neutral/1mk] Increase £200 by 5%.
- (b) [allowed/IB/2mk] After a 5% increase a price is £210. Find the original.
- (c) [allowed/IB/3mk] A quantity rises by 5% then by another 5%. Overall % increase?

T.IV.13.020 · Geometric series in repeating decimals link · core · UK_GCSE_H,IB_AA_SL,IB_AA_HL

- (a) [never/neutral/2mk] n th term of 4, 7, 10, 13, ...
- (b) [never/UK/1mk] 20th term: $a=4, d=3$.
- (c) [never/UK/2mk] S_n for $a=4, d=3$.

T.IV.7.027 · Worded optimisation without calculus (complete square) · challenge · UK_GCSE_H

- (a) [never/neutral/2mk] Factorise $x^2 - 4x + 3$.
- (b) [never/UK/2mk] Solve $x^2 - 4x + 3 = 0$.
- (c) [never/UK/3mk] Complete the square: $x^2 - 2x + 7 = (x-p)^2 + q$. Find p, q .

T.IV.14.022 · Probability with without replacement trees · core · UK_GCSE_H,IB_AA_SL,IB_AI_SL

- (a) [never/neutral/1mk] Bag: 8 red, 5 blue. $P(\text{red})$?
- (b) [never/UK/2mk] Without replacement, $P(\text{both red})$?
- (c) [never/UK/2mk] With replacement, $P(\text{red then blue})$?

T.IV.15.024 · Lurking variables in observational studies · core · AP_STATS,IB_AI_HL,IB_AI_SL

- (a) [allowed/neutral/1mk] $X \sim N(50, 5^2)$. Approx $P(45 < X < 55)$ (empirical rule)?
- (b) [allowed/US/1mk] z-score of $x=60$ ($\mu=50, \sigma=5$)?
- (c) [gdc/US/2mk] GDC: outline finding $P(X > 55)$ for $X \sim N(50, 5^2)$.

T.IV.13.021 · GCSE Higher special sequences (triangular) · strange · UK_GCSE_H

- (a) [never/neutral/2mk] n th term of 6, 9, 12, 15, ...
- (b) [never/UK/1mk] 20th term: $a=6, d=3$.
- (c) [never/UK/2mk] S_n for $a=6, d=3$.

Answer Key — Era IV

Brief solutions. Full reasoning in machine JSON.

T.IV.5.001

- (a) $2\sqrt{2}$ — Extract square factors.
- (b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -2 — Difference of squares.

T.IV.5.002

- (a) $2\sqrt{5}$ — Extract square factors.
- (b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -2 — Difference of squares.

T.IV.5.003

- (a) $3\sqrt{3}$ — Extract square factors.
- (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -1 — Difference of squares.

T.IV.5.004

- (a) $2\sqrt{3}$ — Extract square factors.
- (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -1 — Difference of squares.

T.IV.5.005

- (a) $6\sqrt{2}$ — Extract square factors.
- (b) $\sqrt{2/2}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -2 — Difference of squares.

T.IV.5.006

- (a) $2\sqrt{5}$ — Extract square factors.
- (b) $\sqrt{5/5}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) 1 — Difference of squares.

T.IV.5.007

- (a) $5\sqrt{2}$ — Extract square factors.
- (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) 3 — Difference of squares.

T.IV.5.008

- (a) $2\sqrt{5}$ — Extract square factors.
- (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) 3 — Difference of squares.

T.IV.5.009

- (a) $2\sqrt{5}$ — Extract square factors.
- (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) 3 — Difference of squares.

T.IV.5.010

- (a) $4\sqrt{2}$ — Extract square factors.
- (b) $\sqrt{7/7}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) 3 — Difference of squares.

T.IV.5.011

- (a) $3\sqrt{2}$ — Extract square factors.
- (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -1 — Difference of squares.

T.IV.7.001

- (a) $(x-5)(x-3)$ — Factor pair.
- (b) $x=5$ or $x=3$ — Zero product.
- (c) $p=5$, $q=-18$ — $(x-5)^2 - 25 + 7$.

T.IV.7.002

- (a) $(x-4)(x-7)$ — Factor pair.
- (b) $x=4$ or $x=7$ — Zero product.
- (c) $p=4$, $q=-9$ — $(x-4)^2 - 16 + 7$.

T.IV.7.003

- (a) $(x-4)(x-6)$ — Factor pair.

- (b) $x=4$ or $x=6$ — *Zero product.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.7.004

- (a) $4\sqrt{2}$ — *Extract square factors.*
 (b) $\sqrt{5/5}$ — *Multiply by $\sqrt{b}/\sqrt{b}$.*
 (c) 1 — *Difference of squares.*

T.IV.7.005

- (a) $(x-3)(x-5)$ — *Factor pair.*
 (b) $x=3$ or $x=5$ — *Zero product.*
 (c) $p=3, q=-2$ — $(x-3)^2 - 9 + 7$.

T.IV.7.006

- (a) $(x-2)(x-4)$ — *Factor pair.*
 (b) $x=2$ or $x=4$ — *Zero product.*
 (c) $p=2, q=3$ — $(x-2)^2 - 4 + 7$.

T.IV.7.007

- (a) $(x-5)(x-7)$ — *Factor pair.*
 (b) $x=5$ or $x=7$ — *Zero product.*
 (c) $p=5, q=-18$ — $(x-5)^2 - 25 + 7$.

T.IV.7.008

- (a) $(x-1)(x-6)$ — *Factor pair.*
 (b) $x=1$ or $x=6$ — *Zero product.*
 (c) $p=1, q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.7.009

- (a) $(x-1)(x-4)$ — *Factor pair.*
 (b) $x=1$ or $x=4$ — *Zero product.*
 (c) $p=1, q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.7.010

- (a) $(x-1)(x-2)$ — *Factor pair.*
 (b) $x=1$ or $x=2$ — *Zero product.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.7.011

- (a) $(x-4)(x-6)$ — *Factor pair.*
 (b) $x=4$ or $x=6$ — *Zero product.*
 (c) $p=4, q=-9$ — $(x-4)^2 - 16 + 7$.

T.IV.7.012

- (a) $(x-2)(x-3)$ — *Factor pair.*
 (b) $x=2$ or $x=3$ — *Zero product.*
 (c) $p=2, q=3$ — $(x-2)^2 - 4 + 7$.

T.IV.7.013

- (a) $(x-1)(x-2)$ — *Factor pair.*
 (b) $x=1$ or $x=2$ — *Zero product.*
 (c) $p=1, q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.7.014

- (a) $(x-1)(x-7)$ — *Factor pair.*
 (b) $x=1$ or $x=7$ — *Zero product.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.7.015

- (a) $(x-3)(x-6)$ — *Factor pair.*
 (b) $x=3$ or $x=6$ — *Zero product.*
 (c) $p=3, q=-2$ — $(x-3)^2 - 9 + 7$.

T.IV.7.016

- (a) $(x-4)(x-6)$ — *Factor pair.*
 (b) $x=4$ or $x=6$ — *Zero product.*
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.7.017

- (a) $(x-1)(x-3)$ — *Factor pair.*
 (b) $x=1$ or $x=3$ — *Zero product.*
 (c) $p=1, q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.7.018

- (a) $(x-1)(x-5)$ — Factor pair.
- (b) $x=1$ or $x=5$ — Zero product.
- (c) $p=1$, $q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.7.019

- (a) $(x-2)(x-7)$ — Factor pair.
- (b) $x=2$ or $x=7$ — Zero product.
- (c) $p=2$, $q=3$ — $(x-2)^2 - 4 + 7$.

T.IV.7.020

- (a) $(x-5)(x-2)$ — Factor pair.
- (b) $x=5$ or $x=2$ — Zero product.
- (c) $p=5$, $q=-18$ — $(x-5)^2 - 25 + 7$.

T.IV.7.021

- (a) $(x-4)(x-5)$ — Factor pair.
- (b) $x=4$ or $x=5$ — Zero product.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.7.022

- (a) $(x-4)(x-7)$ — Factor pair.
- (b) $x=4$ or $x=7$ — Zero product.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.7.023

- (a) $(x-2)(x-5)$ — Factor pair.
- (b) $x=2$ or $x=5$ — Zero product.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.1.001

- (a) $x = 7$ — Isolate x .
- (b) $x = 8$ — Expand/divide.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.1.002

- (a) $x = 3$ — Isolate x .
- (b) $x = 4$ — Expand/divide.
- (c) 4, 5, 6, 7 — Integer solutions of inequality.

T.IV.1.003

- (a) $2\sqrt{3}$ — Extract square factors.
- (b) $\sqrt{3/3}$ — Multiply by $\sqrt{b}/\sqrt{b}$.
- (c) -1 — Difference of squares.

T.IV.2.001

- (a) $6x + 36$ — Distribute.
- (b) $x^2 + 7x + 6$ — FOIL.
- (c) $(x+1)(x+6)$ — Factor pair.

T.IV.2.002

- (a) $7x + 21$ — Distribute.
- (b) $x^2 + 9x + 18$ — FOIL.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.2.003

- (a) $7x + 28$ — Distribute.
- (b) $x^2 + 6x + 5$ — FOIL.
- (c) $(x+5)(x+1)$ — Factor pair.

T.IV.2.004

- (a) $2x + 10$ — Distribute.
- (b) $x^2 + 2x + 1$ — FOIL.
- (c) $(x+1)(x+1)$ — Factor pair.

T.IV.2.005

- (a) $4x + 8$ — Distribute.
- (b) $x^2 + 6x + 8$ — FOIL.
- (c) Lab evidence (notes/screenshot) + one verified check — Complete lab steps.

T.IV.2.006

- (a) $6x + 6$ — *Distribute.*
- (b) $x^2 + 11x + 30$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.2.007

- (a) $4x + 20$ — *Distribute.*
- (b) $x^2 + 4x + 3$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.2.008

- (a) $5x + 25$ — *Distribute.*
- (b) $x^2 + 6x + 8$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.2.009

- (a) $6x + 6$ — *Distribute.*
- (b) $x^2 + 7x + 12$ — *FOIL.*
- (c) $(x+4)(x+3)$ — *Factor pair.*

T.IV.2.010

- (a) $5x + 5$ — *Distribute.*
- (b) $x^2 + 7x + 10$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.2.011

- (a) $4x + 4$ — *Distribute.*
- (b) $x^2 + 6x + 9$ — *FOIL.*
- (c) $(x+3)(x+3)$ — *Factor pair.*

T.IV.3.001

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.IV.3.002

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.IV.3.003

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.3.004

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.IV.4.001

- (a) $14 - ad - bc.$
- (b) $[[4,3],[2,5]]$ — *Identity.*
- (c) $(1/14)[[5,-3],[-2,4]] - (1/\det)[[d,-b],[-c,a]].$

T.IV.4.002

- (a) $-1 - ad - bc.$
- (b) $[[1,1],[4,3]]$ — *Identity.*
- (c) $(1/-1)[[3,-1],[-4,1]] - (1/\det)[[d,-b],[-c,a]].$

T.IV.4.003

- (a) $4 - ad - bc.$
- (b) $[[3,1],[5,3]]$ — *Identity.*
- (c) $(1/4)[[3,-1],[-5,3]] - (1/\det)[[d,-b],[-c,a]].$

T.IV.4.004

- (a) $5 - ad - bc.$
- (b) $[[3,1],[1,2]]$ — *Identity.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.4.005

- (a) $-1 - ad - bc.$
- (b) $[[1,4],[1,3]]$ — *Identity.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.6.001

- (a) $7x + 21$ — *Distribute.*
- (b) $x^2 + 2x + 1$ — *FOIL.*
- (c) $(x+1)(x+1)$ — *Factor pair.*

T.IV.6.002

- (a) $2x + 8$ — *Distribute.*
- (b) $x^2 + 6x + 5$ — *FOIL.*
- (c) $(x+5)(x+1)$ — *Factor pair.*

T.IV.6.003

- (a) $7x + 14$ — *Distribute.*
- (b) $x^2 + 5x + 4$ — *FOIL.*
- (c) $(x+1)(x+4)$ — *Factor pair.*

T.IV.6.004

- (a) $3x + 9$ — *Distribute.*
- (b) $x^2 + 9x + 20$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.6.005

- (a) $7x + 7$ — *Distribute.*
- (b) $x^2 + 9x + 20$ — *FOIL.*
- (c) $(x+5)(x+4)$ — *Factor pair.*

T.IV.6.006

- (a) $3x + 15$ — *Distribute.*
- (b) $x^2 + 5x + 4$ — *FOIL.*
- (c) $(x+4)(x+1)$ — *Factor pair.*

T.IV.6.007

- (a) $4x + 16$ — *Distribute.*
- (b) $x^2 + 8x + 16$ — *FOIL.*
- (c) $(x+4)(x+4)$ — *Factor pair.*

T.IV.6.008

- (a) $5x + 20$ — *Distribute.*
- (b) $x^2 + 7x + 12$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.8.001

- (a) $3x + 18$ — *Distribute.*
- (b) $x^2 + 5x + 4$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.8.002

- (a) $4x + 24$ — *Distribute.*
- (b) $x^2 + 6x + 5$ — *FOIL.*
- (c) $(x+1)(x+5)$ — *Factor pair.*

T.IV.8.003

- (a) $5x + 20$ — *Distribute.*
- (b) $x^2 + 9x + 20$ — *FOIL.*
- (c) $(x+5)(x+4)$ — *Factor pair.*

T.IV.8.004

- (a) $7x + 28$ — *Distribute.*
- (b) $x^2 + 4x + 3$ — *FOIL.*
- (c) $(x+3)(x+1)$ — *Factor pair.*

T.IV.8.005

- (a) $6x + 36$ — *Distribute.*
- (b) $x^2 + 5x + 6$ — *FOIL.*
- (c) $(x+2)(x+3)$ — *Factor pair.*

T.IV.9.001

- (a) $2x + 8$ — *Distribute.*
- (b) $x^2 + 10x + 24$ — *FOIL.*
- (c) $(x+4)(x+6)$ — *Factor pair.*

T.IV.9.002

- (a) $4x + 16$ — *Distribute.*
- (b) $x^2 + 7x + 10$ — *FOIL.*
- (c) $(x+5)(x+2)$ — *Factor pair.*

T.IV.9.003

- (a) $5x + 10$ — *Distribute.*
- (b) $x^2 + 6x + 8$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.9.004

- (a) $2x + 10$ — *Distribute.*
- (b) $x^2 + 6x + 8$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.9.005

- (a) $3x + 9$ — *Distribute.*
- (b) $x^2 + 7x + 10$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.9.006

- (a) $6x + 12$ — *Distribute.*
- (b) $x^2 + 7x + 12$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.10.001

- (a) $x = 9$ — *Isolate x.*
- (b) $x = 10$ — *Expand/divide.*
- (c) 10, 11, 12, 13 — *Integer solutions of inequality.*

T.IV.11.001

- (a) $x = 3$ — *Isolate x.*
- (b) $x = 4$ — *Expand/divide.*
- (c) 4, 5, 6, 7 — *Integer solutions of inequality.*

T.IV.11.002

- (a) $5\sqrt{2}$ — *Extract square factors.*
- (b) $\sqrt{2/2}$ — *Multiply by $\sqrt{b}/\sqrt{b}$.*
- (c) -2 — *Difference of squares.*

T.IV.11.003

- (a) $5\sqrt{3}$ — *Extract square factors.*
- (b) $\sqrt{3/3}$ — *Multiply by $\sqrt{b}/\sqrt{b}$.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.12.001

- (a) £126 — $\times 1.05$.
- (b) £120 — $\div 1.05$.
- (c) 10.25% — *Multiply multipliers; convert to %.*

T.IV.12.002

- (a) £132 — $\times 1.1$.
- (b) £120 — $\div 1.1$.
- (c) 21.00% — *Multiply multipliers; convert to %.*

T.IV.12.003

- (a) £134.40 — $\times 1.12$.
- (b) £119.64 — $\div 1.12$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.12.004

- (a) £132 — $\times 1.1$.
- (b) £120 — $\div 1.1$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.12.005

- (a) £300 — $\times 1.2$.
- (b) £250 — $\div 1.2$.
- (c) 44.00% — *Multiply multipliers; convert to %.*

T.IV.12.006

- (a) $\pounds 287.50 - \times 1.15$.
 (b) $\pounds 249.57 - \div 1.15$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.12.007

- (a) $\pounds 250 - \times 1.25$.
 (b) $\pounds 200 - \div 1.25$.
 (c) 56.25% — *Multiply multipliers; convert to %*.

T.IV.12.008

- (a) $\pounds 89.60 - \times 1.12$.
 (b) $\pounds 79.46 - \div 1.12$.
 (c) 25.44% — *Multiply multipliers; convert to %*.

T.IV.12.009

- (a) $\pounds 300 - \times 1.2$.
 (b) $\pounds 250 - \div 1.2$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.12.010

- (a) $\pounds 138 - \times 1.15$.
 (b) $\pounds 120 - \div 1.15$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.12.011

- (a) $\pounds 420 - \times 1.05$.
 (b) $\pounds 400 - \div 1.05$.
 (c) 10.25% — *Multiply multipliers; convert to %*.

T.IV.12.012

- (a) $\pounds 300 - \times 1.2$.
 (b) $\pounds 250 - \div 1.2$.
 (c) 44.00% — *Multiply multipliers; convert to %*.

T.IV.12.013

- (a) $\pounds 448 - \times 1.12$.
 (b) $\pounds 400 - \div 1.12$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.12.014

- (a) $\pounds 262.50 - \times 1.05$.
 (b) $\pounds 249.52 - \div 1.05$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.12.015

- (a) $\pounds 312.50 - \times 1.25$.
 (b) $\pounds 249.6 - \div 1.25$.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.13.001

- (a) $6n + 1$ — *Difference 6*.
 (b) $121 - a + (n-1)d$.
 (c) $340 - n/2[2a + (n-1)d]$.

T.IV.13.002

- (a) $2n$ — *Difference 2*.
 (b) $40 - a + (n-1)d$.
 (c) $110 - n/2[2a + (n-1)d]$.

T.IV.13.003

- (a) $6n + -1$ — *Difference 6*.
 (b) $119 - a + (n-1)d$.
 (c) $320 - n/2[2a + (n-1)d]$.

T.IV.13.004

- (a) $6n + 3$ — *Difference 6*.
 (b) $123 - a + (n-1)d$.
 (c) $360 - n/2[2a + (n-1)d]$.

T.IV.13.005

- (a) $6n + -2$ — *Difference 6*.
 (b) $118 - a + (n-1)d$.

(c) $310 - n/2[2a+(n-1)d]$.

T.IV.13.006

- (a) $5n + 4$ — *Difference 5.*
(b) $104 - a+(n-1)d$.
(c) $315 - n/2[2a+(n-1)d]$.

T.IV.13.007

- (a) $5n + 2$ — *Difference 5.*
(b) $102 - a+(n-1)d$.
(c) $295 - n/2[2a+(n-1)d]$.

T.IV.13.008

- (a) $6n$ — *Difference 6.*
(b) $120 - a+(n-1)d$.
(c) $330 - n/2[2a+(n-1)d]$.

T.IV.13.009

- (a) $4n + 3$ — *Difference 4.*
(b) $83 - a+(n-1)d$.
(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.13.010

- (a) $7n + -3$ — *Difference 7.*
(b) $137 - a+(n-1)d$.
(c) $355 - n/2[2a+(n-1)d]$.

T.IV.13.011

- (a) $2n + 3$ — *Difference 2.*
(b) $43 - a+(n-1)d$.
(c) $140 - n/2[2a+(n-1)d]$.

T.IV.13.012

- (a) Base case; inductive hypothesis; inductive step (+ conclusion) — *Standard structure.*
(b) $LHS=k(k+1)/2+(k+1)=(k+1)(k+2)/2$ — *Add $k+1$; factor.*
(c) e.g. missing base case / assuming the result / no conclusion — *Structure marks.*

T.IV.13.013

- (a) $7n + -1$ — *Difference 7.*
(b) $139 - a+(n-1)d$.
(c) $375 - n/2[2a+(n-1)d]$.

T.IV.13.014

- (a) $2n$ — *Difference 2.*
(b) $40 - a+(n-1)d$.
(c) $110 - n/2[2a+(n-1)d]$.

T.IV.13.015

- (a) $2n + 1$ — *Difference 2.*
(b) $41 - a+(n-1)d$.
(c) $120 - n/2[2a+(n-1)d]$.

T.IV.13.016

- (a) $4n + -2$ — *Difference 4.*
(b) $78 - a+(n-1)d$.
(c) $200 - n/2[2a+(n-1)d]$.

T.IV.13.017

- (a) $7n + -4$ — *Difference 7.*
(b) $136 - a+(n-1)d$.
(c) $345 - n/2[2a+(n-1)d]$.

T.IV.14.001

- (a) $5/7$ — $5/7$.
(b) $10/21$ — *Product along tree.*
(c) $10/49$ — *Independent draws.*

T.IV.14.002

- (a) $5/9$ — $5/9$.
(b) $5/18$ — *Product along tree.*
(c) $20/81$ — *Independent draws.*

T.IV.14.003

- (a) $4/9$ — $4/9$.
- (b) $1/6$ — *Product along tree.*
- (c) $20/81$ — *Independent draws.*

T.IV.14.004

- (a) $7/13$ — $7/13$.
- (b) $7/26$ — *Product along tree.*
- (c) $42/169$ — *Independent draws.*

T.IV.14.005

- (a) $3/4$ — $6/8$.
- (b) $15/28$ — *Product along tree.*
- (c) $3/16$ — *Independent draws.*

T.IV.14.006

- (a) $7/11$ — $7/11$.
- (b) $21/55$ — *Product along tree.*
- (c) $28/121$ — *Independent draws.*

T.IV.14.007

- (a) $2/3$ — $8/12$.
- (b) $14/33$ — *Product along tree.*
- (c) $2/9$ — *Independent draws.*

T.IV.14.008

- (a) $4/7$ — $4/7$.
- (b) $2/7$ — *Product along tree.*
- (c) $12/49$ — *Independent draws.*

T.IV.14.009

- (a) $7/13$ — $7/13$.
- (b) $7/26$ — *Product along tree.*
- (c) $42/169$ — *Independent draws.*

T.IV.14.010

- (a) $1/2$ — $4/8$.
- (b) $3/14$ — *Product along tree.*
- (c) $1/4$ — *Independent draws.*

T.IV.14.011

- (a) $4/7$ — $8/14$.
- (b) $4/13$ — *Product along tree.*
- (c) $12/49$ — *Independent draws.*

T.IV.14.012

- (a) $4/5$ — $8/10$.
- (b) $28/45$ — *Product along tree.*
- (c) $4/25$ — *Independent draws.*

T.IV.14.013

- (a) $1/2$ — $7/14$.
- (b) $3/13$ — *Product along tree.*
- (c) $1/4$ — *Independent draws.*

T.IV.14.014

- (a) The governing definition/theorem for Conditional probability medical screening FRQ — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.IV.14.015

- (a) $7/11$ — $7/11$.
- (b) $21/55$ — *Product along tree.*
- (c) $28/121$ — *Independent draws.*

T.IV.14.016

- (a) $8/13$ — $8/13$.
- (b) $14/39$ — *Product along tree.*
- (c) $40/169$ — *Independent draws.*

T.IV.14.017

- (a) The governing definition/theorem for Conditional probability FRQ-style 1 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.IV.14.018

- (a) The governing definition/theorem for Conditional probability FRQ-style 2 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.IV.14.019

- (a) The governing definition/theorem for Conditional probability FRQ-style 3 — *Name it precisely.*
- (b) Because [hypothesis], by [theorem], therefore [claim]. — *FRQ communication.*
- (c) Accept correct on-skill calculation + interpretation — *Show working + meaning.*

T.IV.15.001

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.002

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.003

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.004

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.005

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.006

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.007

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.008

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.009

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.010

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.011

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.
- (c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.012

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.013

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.014

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.015

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.016

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.017

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.018

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.019

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.15.020

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.15.021

(a) $\approx 68\%$ — *1 SD rule.*

(b) $2 - (x - \mu)/\sigma$.

(c) $\text{normalcdf}(\mu + \sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.5.012

(a) $2\sqrt{2}$ — *Extract square factors.*

(b) $\sqrt{2}/2$ — *Multiply by $\sqrt{b}/\sqrt{b}$.*

(c) -2 — *Difference of squares.*

T.IV.5.013

(a) $3\sqrt{2}$ — *Extract square factors.*

(b) $\sqrt{2}/2$ — *Multiply by $\sqrt{b}/\sqrt{b}$.*

(c) -2 — *Difference of squares.*

T.IV.7.024

(a) $(x-3)(x-5)$ — *Factor pair.*

(b) $x=3$ or $x=5$ — *Zero product.*

(c) $p=3, q=-2$ — $(x-3)^2 - 9 + 7$.

T.IV.7.025

(a) $(x-4)(x-5)$ — *Factor pair.*

(b) $x=4$ or $x=5$ — *Zero product.*

(c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.7.026

(a) $(x-5)(x-3)$ — *Factor pair.*

(b) $x=5$ or $x=3$ — *Zero product.*

(c) $p=5, q=-18$ — $(x-5)^2 - 25 + 7$.

T.IV.2.012

- (a) $3x + 6$ — *Distribute*.
 (b) $x^2 + 9x + 20$ — *FOIL*.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.2.013

- (a) $4x + 16$ — *Distribute*.
 (b) $x^2 + 6x + 5$ — *FOIL*.
 (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps*.

T.IV.6.009

- (a) $5x + 25$ — *Distribute*.
 (b) $x^2 + 2x + 1$ — *FOIL*.
 (c) $(x+1)(x+1)$ — *Factor pair*.

T.IV.6.010

- (a) $7x + 14$ — *Distribute*.
 (b) $x^2 + 8x + 12$ — *FOIL*.
 (c) $(x+2)(x+6)$ — *Factor pair*.

T.IV.12.016

- (a) £287.50 — $\times 1.15$.
 (b) £249.57 — $\div 1.15$.
 (c) 32.25% — *Multiply multipliers; convert to %*.

T.IV.12.017

- (a) £480 — $\times 1.2$.
 (b) £400 — $\div 1.2$.
 (c) 44.00% — *Multiply multipliers; convert to %*.

T.IV.13.018

- (a) $6n + -1$ — *Difference 6*.
 (b) $119 - a + (n-1)d$.
 (c) $320 - n/2[2a + (n-1)d]$.

T.IV.13.019

- (a) $6n$ — *Difference 6*.
 (b) $120 - a + (n-1)d$.
 (c) $330 - n/2[2a + (n-1)d]$.

T.IV.14.020

- (a) $3/5 - 6/10$.
 (b) $1/3$ — *Product along tree*.
 (c) $6/25$ — *Independent draws*.

T.IV.14.021

- (a) $1/2 - 4/8$.
 (b) $3/14$ — *Product along tree*.
 (c) $1/4$ — *Independent draws*.

T.IV.15.022

- (a) $\approx 68\%$ — *1 SD rule*.
 (b) $2 - (x-\mu)/\sigma$.
 (c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf*.

T.IV.15.023

- (a) $\approx 68\%$ — *1 SD rule*.
 (b) $2 - (x-\mu)/\sigma$.
 (c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf*.

T.IV.4.006

- (a) $0 - ad - bc$.
 (b) $[[4,5],[4,5]]$ — *Identity*.
 (c) No inverse — $(1/\det)[[d,-b],[-c,a]]$.

T.IV.8.006

- (a) $6x + 18$ — *Distribute*.
 (b) $x^2 + 11x + 30$ — *FOIL*.
 (c) $(x+5)(x+6)$ — *Factor pair*.

T.IV.9.007

- (a) $2x + 8$ — *Distribute.*
- (b) $x^2 + 2x + 1$ — *FOIL.*
- (c) Lab evidence (notes/screenshot) + one verified check — *Complete lab steps.*

T.IV.3.005

- (a) Top view (from above) — *Plan = bird's-eye.*
- (b) 6 — *Six faces.*
- (c) Isosceles triangle with base L — *Front sees triangular profile.*

T.IV.1.004

- (a) $x = 1$ — *Isolate x .*
- (b) $x = 2$ — *Expand/divide.*
- (c) 2, 3, 4, 5 — *Integer solutions of inequality.*

T.IV.11.004

- (a) $3\sqrt{3}$ — *Extract square factors.*
- (b) $\sqrt[7]{7}$ — *Multiply by $\sqrt[7]{b}/\sqrt[7]{b}$.*
- (c) 3 — *Difference of squares.*

T.IV.12.018

- (a) £210 — $\times 1.05$.
- (b) £200 — $\div 1.05$.
- (c) 10.25% — *Multiply multipliers; convert to %.*

T.IV.13.020

- (a) $3n + 1$ — *Difference 3.*
- (b) $61 - a + (n-1)d$.
- (c) $175 - n/2[2a + (n-1)d]$.

T.IV.7.027

- (a) $(x-1)(x-3)$ — *Factor pair.*
- (b) $x=1$ or $x=3$ — *Zero product.*
- (c) $p=1, q=6$ — $(x-1)^2 - 1 + 7$.

T.IV.14.022

- (a) $8/13$ — $8/13$.
- (b) $14/39$ — *Product along tree.*
- (c) $40/169$ — *Independent draws.*

T.IV.15.024

- (a) $\approx 68\%$ — *1 SD rule.*
- (b) $2 - (x-\mu)/\sigma$.
- (c) $\text{normalcdf}(\mu+\sigma, \infty, \mu, \sigma) \approx 16\%$ — *Right-tail normalcdf.*

T.IV.13.021

- (a) $3n + 3$ — *Difference 3.*
- (b) $63 - a + (n-1)d$.
- (c) $195 - n/2[2a + (n-1)d]$.